

ARDHRA.K

AI ML ENGINEER

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SUMMARY

Machine Learning and AI Developer with strong foundation in designing, developing, and deploying ML/DL models across Computer Vision, NLP, and predictive analytics. Skilled in building CNNs, working with Transformers and LLMs, and optimizing model performance through hyperparameter tuning and evaluation. Proficient in Python, TensorFlow, PyTorch, scikit-learn, and OpenCV. Adept at data preprocessing, feature engineering, and delivering scalable, high-impact solutions using modern ML frameworks and tools. Experienced in end-to-end machine learning pipelines, from data collection and preprocessing to model deployment and performance monitoring in real-world applications.

Education

Bachelor of Computer Applications (BCA), SNDP College, Perinthalmanna CGPA: 7.35 / 10.0	2025
Class 12th, Kerala Board of Public Examination Percentage: 79%	2022
Class 10th, Central Board of Secondary Education Percentage: 78.6%	2020

skills

Languages: Python, SQL

ML / DL Frameworks: scikit-learn, TensorFlow, Keras, PyTorch, OpenCV

Core ML: Supervised learning, CNNs, hyperparameter tuning, model evaluation (precision, recall, F1, ROC-AUC, confusion matrix)

Data Handling: Pandas, NumPy, data cleaning & preprocessing, EDA, feature engineering

Visualization & BI: Matplotlib, Seaborn, Power BI, Tableau

Tools: Jupyter Notebook, Git, MongoDB, Flask

Exploring: NLP, Transformers, Hugging Face, LangChain, RAG (self-study, not yet project-tested)

PROJECT

Handwritten Digit Recognition (CNN, TensorFlow/Keras)

- Preprocessed 70,000 MNIST images using normalization, reshaping, and one-hot encoding.
- Designed a CNN with convolutional, pooling, dropout, and fully connected layers; tuned epochs, batch size, filters, and kernel size.
- Reached 98-99% test accuracy; evaluated with precision, recall, F1-score, and confusion matrix, then built a real-time digit classifier for user-drawn input.
- Ran error analysis on misclassified samples to identify and reduce systematic model errors.

Liver Patient Multi Model Prediction (Ensemble Models, scikit-learn)

- Cleaned a 582-record medical dataset: mean imputation, deduplication, and label encoding.
- Compared Random Forest, Gradient Boosting, and XGBoost; tuned Gradient Boosting via Random Search.
- Selected Recall as the priority metric for this medical use case, achieving 100% detection of positive cases (Accuracy 0.71, Precision 0.71, F1 0.83).
- Generated confusion matrices and classification reports to support the final model recommendation.

Flight Price Analysis and Prediction (Regression, scikit-learn)

- Cleaned and feature-engineered flight fare data using Pandas and NumPy; ran EDA with Matplotlib and Seaborn to surface pricing trends.
- Built and compared regression models, evaluating with R^2 , MAE, and RMSE to derive fare-pattern insights.